Hirtz & Carter

Project Proposal – Automated Image Quality Assurance (Image-QA) System

Industry: Retail (e-commerce)
Date: 15 Jan 2026

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1. Executive Summary

Item	Summary
Business Challenge	Hirtz & Carter manually reviews thousands of product images against detailed QA checklists (background, colour accuracy, retouching quality, compliance, etc.). The process is time-consuming, error-prone, and creates bottlenecks that delay product publishing.
Proposed Solution	Deploy a **Clarifai-powered multimodal AI pipeline** that (1) automatically evaluates each image against the retail QA criteria, (2) returns a structured JSON verdict with explanations, and (3) integrates seamlessly into the existing DAM / CMS workflow. The core engine uses a **Vision-Language Model (VLM)** for high-level reasoning and a **lightweight detection model** for any visual-only checks that require explicit localization (e.g., background-color deviation, presence of shadows).
Key Value	• 70-80 % reduction in manual QA effort (estimated ROI ≈ 12 months). • Consistent, auditable decisions with textual explanations for compliance teams. • Scalable cloud-managed inference – no on-prem GPU procurement.

Item	Summary
Outcomes	• Faster time-to-market (average image-to-publish latency < 2 s). • Lower re-work cost (≈ 30 % drop in retouching hours). • Ability to enforce marketplace-specific image standards automatically.

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2. Problem Statement

1. **Manual, labor-intensive QA** – Reviewers must read long checklists and manually flag defects, leading to inconsistent judgments.
2. **High volume & growth** – Current catalog ≈ 150 k images/month; projected 30 % YoY growth will outpace human capacity.
3. **Compliance risk** – Missed image violations (e.g., wrong background, colour shift) cause marketplace penalties and lost sales.
4. **Lack of analytics** – No systematic capture of defect types or root-cause data, making process improvement difficult.

Quantified impact (estimates from the discovery phase):

Metric	Current	Desired	Business Impact
Images reviewed per FTE per day	250	500 (automation)	+2 FTE capacity
Average time to approve an image	45 s	≤ 10 s	Faster publishing
Re-work cost per month	$22 k	$15 k	$7 k savings
Marketplace rejection rate	2.4 %	< 0.5 %	Revenue protection

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3. Proposed Solution

3.1 Overview

A **two-stage AI pipeline** hosted on Clarifai Cloud:

1. **Stage 1 – Vision-Language Reasoning (VLM)**
   Model: `mm-poly-8b` (frontier multimodal model).
   Function: Reads the image, interprets it against the textual QA checklist, and returns a **structured JSON** with:
   - `overall_status` (PASS/FAIL)
   - `defect_list` (array of detected issues)
   - `explanation` (human-readable rationale)

2. **Stage 2 – Targeted Detection (optional)**
   Model(s):
   - **Object Detection** – `clarifai-retail-bg-detector` (detects background colour, shadows, stray objects).
   - **Classification** – `clarifai-color-accuracy-cls` (verifies colour consistency).
   These models output **bounding boxes / confidence scores** when a visual localization is required (e.g., “shadow present”).

3. **Orchestration via Clarifai Workflows** – Sequential execution, branching if VLM flags a “requires visual localization” condition.

4. **Result Delivery** – JSON verdict is sent back to the DAM/ CMS via a **REST webhook**; a lightweight UI dashboard shows per-image status and aggregate statistics.

3.2 Why This Architecture

Requirement	How it’s satisfied
	VLM (`mm-poly-8b`) excels at answering open-ended visual questions in text.

Requirement	How it's satisfied
Complex visual reasoning (e.g., "does the lighting look natural?")	
Explainability (need for textual justification)	VLM returns a natural-language explanation; detection models supply visual evidence when needed.
Scalable throughput (up to 300 imgs/min peak)	Cloud-managed GPU pool; auto-scales the workflow.
Compliance & audit	Structured JSON + optional bounding boxes provide traceable evidence.
Future-proofing	Model Evaluation Phase lets H&C benchmark 2-3 alternative VLMs (e.g., `open-clip-v2`, `LLaVA-1.5-8b`) before final lock-in.
Cost efficiency	Use VLM for high-level checks (single inference per image) and lightweight detection only when needed, reducing total GPU seconds.

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4. Technical Approach

4.1 Data Requirements & Pre-processing

Step	Description
Ingestion	Images pulled from existing DAM via REST API or S3 bucket trigger.
Normalization	Convert to RGB, resize to max 1024 px (preserves quality, meets VLM input limits).

Step	Description
Metadata Enrichment	Attach SKU, source, and any existing manual QA label (if available).
Quality Gate	Validate file integrity (corruption, colour profile) before inference.

4.2 Model Selection

Model	Type	Primary Use	Expected Latency (GPU = Standard tier)
`mm-poly-8b`	VLM (frontier)	Textual QA reasoning, classification, explanation	~1.2 s per 1024×1024 image
`clarifai-retail-bg-detector`	Object Detection	Detect background colour deviations, shadows, stray objects	~0.4 s
`clarifai-color-accuracy-cls`	Image Classification	Verify colour match against reference swatches	~0.2 s

Model Evaluation Phase (Weeks 1-3) will benchmark these against two alternatives (e.g., `LLaVA-1.5-8b`, `open-clip-v2`) using a **hold-out set of 5 k labeled images**

from H&C. Decision criteria: accuracy ≥ 90 % on critical defects, latency ≤ 2 s, cost ≤ $0.005 per inference.

4.3 Workflow Design

```
flowchart TD
    A[Image Ingest (DAM/S3)] --> B[Pre-process (Resize, Validate)]
    B --> C[Stage-1: VLM (mm-poly-8b)]
    C --> D{VLM Flags Visual Defect?}
    D -->|Yes| E[Stage-2: Detection Models]
    D -->|No| F[Return PASS JSON]
    E --> G[Aggregate Results]
    G --> H[Webhook / Dashboard]
    F --> H
```

- **Parallelism:** Multiple images are processed concurrently (max batch size = 8) on a shared GPU pool.
- **Error handling:** Failed inference routes to a dead-letter queue; admins receive an alert.

4.4 Integration

Component	Integration Method
DAM / CMS	REST webhook (POST JSON) or direct API call from Clarifai workflow.
Authentication	Personal Access Token (PAT) – stored in H&C's secret manager.
SDK	Python SDK (`pip install clarifai`) for any custom scripts (e.g., bulk re-scan).
OpenAI-compatible endpoint	If H&C already uses OpenAI clients, they can call `https://api.clarifai.com/v2/ext/openai/v1/chat/completions` with VLM prompts.

Component	Integration Method
Dashboard	Optional lightweight React UI hosted on Vercel, consuming the webhook payload.

4.5 GPU Reservation Recommendation

Estimate	Details
Peak throughput	300 imgs/min → 5 imgs/sec
GPU type	**Standard tier** (NVIDIA A10 or equivalent) – provides ~2 imgs/sec per GPU for `mm-poly-8b`.
GPU count	**3 GPUs** (auto-scale to 5 % headroom).
Commitment	12-month reserved capacity (≈3 GPU-months).
Annual cost (Clarifai managed)	Approx. **$24,000** (incl. support, monitoring).
Pay-as-you-go fallback	$0.008 per GPU-second if usage exceeds reservation.

> ***Note:*** *If the final throughput requirement drops below 150 imgs/min, a* ***single GPU*** *on the* ***Entry tier*** *could be sufficient, cutting cost by ~40 %.*

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5. Deliverables

#	Deliverable	Acceptance Criteria
1	**Solution Architecture Document** (incl. mermaid diagram, data flow)	Reviewed & signed-off by H&C technical lead.
2		

#	Deliverable	Acceptance Criteria
	Clarifai Workflow (VLM + detection orchestration) deployed to a staging environment.	Executes end-to-end on a sample batch of 100 images with <2 s latency per image.
3	**Integration Code** (Python SDK wrappers, webhook endpoint, optional UI)	Successfully receives images from DAM and posts JSON back without errors.
4	**Model Evaluation Report** (benchmark tables, recommendation)	Demonstrates ≥90 % defect detection accuracy on the validation set.
5	**User Guide & SOP** (how to trigger scans, interpret results, handle failures)	Documentation reviewed by H&C QA team; test-run completed.
6	**Monitoring & Alerting Dashboard** (Grafana/Clarifai Insights)	Shows real-time throughput, error rates, and KPI trends.
7	**Project Closure Report** (final metrics, ROI estimate, next-step roadmap)	All milestones met, sign-off obtained.

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6. Timeline & Milestones

Phase	Duration	Key Activities	Milestone
Kick-off & Requirements	1 wk	Align on data access, QA checklist mapping	Signed Requirements Doc
Data Prep & Labelling	2 wks	Ingest historical images, clean labels, create validation set	Validation Set Ready
	3 wks		

Phase	Duration	Key Activities	Milestone
Model Evaluation Phase		Benchmark `mm-poly-8b` + 2 alternatives, select detection models	Evaluation Report
Workflow Development	2 wks	Build Clarifai workflow, add preprocessing, integrate webhook	Staging Workflow Live
Integration & UI	2 wks	Connect to DAM/CMS, develop dashboard, implement auth	End-to-End Demo
Testing & Performance Tuning	1 wk	Load-test at peak throughput, latency optimization	Performance Sign-off
User Acceptance Testing (UAT)	1 wk	H&C QA team runs real-world batch, provides feedback	UAT Sign-off
Production Deployment	1 wk	Reserve GPUs, switch DNS/webhook to prod, enable monitoring	Go-Live
Post-Launch Support	4 weeks	Hyper-care, bug fixes, KPI reporting	Project Closure

Total Project Duration: ≈ 12 weeks (3 months)

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7. Pricing Considerations

Item	Cost (USD)	Notes
	$24,000	

Item	Cost (USD)	Notes
GPU Reservation (3 × Standard tier, 12 mo)		Managed by Clarifai; includes 99.9 % SLA.
Professional Services (design, integration, UAT)	$18,000	120 hrs @ $150/hr (incl. architecture, workflow dev, docs).
Support & Maintenance (first 3 months)	$6,000	24/7 ticket support, performance monitoring.
Optional Add-Ons		
• Custom Model Fine-tuning (if VLM under-performs)	$5,000 – $12,000 (depends on data volume)	Only after evaluation phase.
• On-prem Deployment (if required by compliance)	TBD	Would need self-hosted GPU hardware.

Total Estimated First-Year Investment: ≈ $48,000 (incl. GPU reservation, services, support).

Potential ROI: If manual QA cost is $22 k/month, a 30 % reduction saves **$79 k** in the first year, delivering a **~1.6 ×** return on investment.

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8. Risks & Mitigations

Risk	Impact	Likelihood	Mitigation
Insufficient labeled data for evaluation	May delay model selection or lower accuracy.	Medium	Use semi-supervised labeling, synthetic augmentation, and active learning loops.

Risk	Impact	Likelihood	Mitigation
Latency exceeds threshold during peak load	Could stall publishing pipeline.	Low-Medium	Auto-scale GPU pool; fallback to batch mode for overflow; pre-warm models.
Data privacy / compliance block (e.g., GDPR)	Inability to send images to cloud.	Low	Offer a **private VPC endpoint** or on-prem inference container (if required).
Model drift (new product styles)	Degraded accuracy over time.	Medium	Implement continuous monitoring; schedule quarterly re-evaluation with fresh data.
Integration complexity with legacy DAM	Longer implementation time.	Medium	Provide adapter layer (simple Python script) and detailed API spec; allocate extra integration hours.
Stakeholder resistance to automated decisions	Low adoption.	Low	Provide clear explanations, keep human-in-the-loop for “FAIL” cases, run pilot with KPI dashboards.

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9. Success Metrics

KPI	Baseline (pre-project)	Target (post-deployment)	Measurement Method
Manual QA time per image	45 s	≤ 10 s	

KPI	Baseline (pre-project)	Target (post-deployment)	Measurement Method
			Log timestamps from DAM workflow.
Pass-rate after AI review	85 % (human-verified)	95 % (AI-approved)	Compare AI verdict vs. human audit on random sample.
Defect detection accuracy	N/A	≥90 % on critical defects (background, colour, shadows)	Confusion matrix on validation set.
Re-work cost reduction	$22 k/mo	≤$15 k/mo	Finance tracking of retouching hours.
Throughput	150 imgs/min (manual)	300 imgs/min (auto)	System logs of processed images.
Compliance violations (marketplace)	2.4 % of listings	<0.5 %	Marketplace audit reports.

Monthly KPI dashboards will be shared with H&C leadership.

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10. Next Steps

Action	Owner	Due Date
Kick-off meeting – confirm stakeholder list, data access, and success criteria.	H&C Project Lead	22 Jan 2026

Action	Owner	Due Date
Provide sample data – 5 k labeled images (or raw images + QA logs).	H&C Data Owner	24 Jan 2026
Sign Statement of Work (SOW) and issue PAT for Clarifai API.	H&C Legal / IT	27 Jan 2026
Reserve GPU capacity (initial 1-GPU for dev).	Clarifai Account Manager	28 Jan 2026
Begin Phase 1 (Data Prep & Labelling)	Clarifai Engineering	29 Jan 2026
Schedule weekly status sync (30 min).	Both parties	Every Friday

Once the above items are completed, we will commence the **Model Evaluation Phase** (Week 1-3) and deliver the first **Proof-of-Concept** within 4 weeks.

Thank you

We appreciate the opportunity to partner with **Hirtz & Carter** on modernizing your image quality workflow. Our team is ready to deliver a robust, explainable AI solution that scales with your catalog growth and protects your brand reputation.

Please let us know if any section requires further detail or adjustment.

Clarifai Platform Capabilities

Full documentation: https://docs.clarifai.com/

Key Platform Features

Capability	Description	Documentation
Model Library	1000+ pre-built models (VLMs, LLMs, CV, Audio)	clarifai.com/explore
Workflows	Chain multiple models in pipelines	docs.clarifai.com/create/workflows
AI Agents	Autonomous agents with tool calling	docs.clarifai.com/compute/agents
Vector Search	Semantic search for RAG applications	docs.clarifai.com/create/search
Pipelines	Async, long-running MLOps workflows	docs.clarifai.com/compute/pipelines
Compute Orchestration	Deploy anywhere with auto-scaling	docs.clarifai.com/compute/overview
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Model Selection Strategy

Vision-Language Models (VLMs) vs Legacy CV Models

Modern vision-language models significantly outperform traditional computer vision models for most enterprise use cases. We recommend VLMs as the default approach:

Capability	Vision-Language Models (VLMs)	Legacy CV Models
Understanding	Comprehends image context and meaning	Pattern matching only

Capability	Vision-Language Models (VLMs)	Legacy CV Models
Reasoning	Can explain WHY something was detected	Binary yes/no output
Flexibility	Adapt via natural language prompts	Requires retraining
Edge Cases	Handles novel situations naturally	Fails on unseen patterns
Deployment	Days (no training needed)	Weeks to months
Maintenance	Update prompts as needed	Retrain for any change

Recommended Models

Use Case	Recommended Model	Parameters	GPU Tier
Visual Understanding	mm-poly-8b *(Clarifai)*	8B	Standard (24GB)
Document Analysis	Llama 3.2 Vision	11B	Standard (24GB)
Complex Reasoning	Qwen2-VL	7-72B	Standard to Enterprise
Cost-Sensitive	Phi-3 Vision	4.2B	Entry (16GB)
High Accuracy	LLaVA 1.6	7-34B	Standard to Performance

Model Size → GPU Requirements

When selecting a model, the parameter count determines the minimum GPU VRAM needed:

Model Size	VRAM (FP16)	VRAM (INT8)	Recommended GPU Tier	Example Models
1-3B	6-8 GB	3-4 GB	Entry (16GB)	Small VLMs, embedders
7-8B	14-16 GB	7-8 GB	Entry (16GB) or Standard (24GB)	Llama 3.1 8B, Mistral 7B
13-14B	26-28 GB	13-14 GB	Standard (24GB) or Performance (48GB)	Llama 2 13B
30-34B	60-70 GB	30-35 GB	Performance (48GB) or Enterprise (80GB)	CodeLlama 34B
70B	140 GB	70 GB	Enterprise (80GB) or Multi-GPU	Llama 3.1 70B
100B+	200+ GB	100+ GB	Multi-GPU (2-8x H100)	Large proprietary models

Notes: - FP16 (half precision) is standard for quality inference - INT8 quantization reduces VRAM ~50% with minimal quality loss - Add 20-30% overhead for KV cache during inference - Clarifai's GPU fractioning allows multiple smaller models on one GPU

Model Evaluation Phase (Recommended)

Before finalizing the architecture, we recommend a **2-week Model Evaluation Phase**:

1. **Curate Evaluation Dataset** (100-500 representative samples)
2. **Benchmark VLMs**: - Start with Clarifai's mm-poly-8b for vision tasks - Test 2-3 alternative VLMs for comparison - Evaluate accuracy, latency, and cost tradeoffs
3. **Measure Key Metrics**: - Accuracy on your specific use cases - Latency per request - Cost per 1,000 predictions
4. **Document Results** and select best model per use case

5. **Only consider custom training** if VLMs cannot meet requirements

When Custom Training May Be Needed

Custom model training should only be considered when: - VLMs cannot achieve required accuracy after prompt optimization - Latency requirements are <20ms per prediction - Throughput requirements exceed 500 images/second - Offline/edge deployment is mandatory - Cost optimization at massive scale (after validation)

GPU Reservation Recommendation

Clarifai Compute Orchestration handles all infrastructure. You simply reserve the GPU capacity you need.

Compute Orchestration Benefits (https://docs.clarifai.com/compute/overview)

- **Auto-scaling**: Scale from zero to infinity based on demand
- **GPU Fractioning**: Run multiple models per GPU for efficiency
- **Model Packing**: Up to 3.7x compute efficiency
- **Cost Savings**: 60-90% savings possible vs. raw cloud GPUs
- **Zero Management**: Clarifai handles all infrastructure

Estimated GPU Requirements

Based on your project scope, we recommend:

Phase	GPUs	Tier	Monthly Cost	Use Case
Pilot	1-2	Standard (24GB)	~$750-1,500	Initial development and testing
Production	2-4	Standard/ Performance	~$1,500-5,000	Production inference workloads
Scale	4+		~$5,000+	

Phase	GPUs	Tier	Monthly Cost	Use Case
		Performance (48GB)		High-volume production

GPU Tier Guide

Tier	Annual Cost	Best For
Entry (16GB)	~$5,000-6,000	Development, testing, light inference
Standard (24GB)	~$8,000-13,000	Production inference
Performance (48GB)	~$18,000-25,000	LLM inference, high throughput
Enterprise (80GB+)	~$21,000-65,000	Large models, enterprise scale

What You Get

- **Guaranteed GPU availability** - No capacity concerns
- **Predictable annual costs** - Fixed pricing for budgeting
- **Zero infrastructure management** - Clarifai handles everything
- **Dedicated support** - Solution Engineering assistance

Integration Approach

Clarifai API (https://docs.clarifai.com/compute/inference/)

Integration is straightforward using Clarifai's managed API:

```
from clarifai.client import Model

# Select a model from clarifai.com/explore
model = Model(url="https://clarifai.com/user/app/models/model-id")
```

```
# Make predictions
response = model.predict(inputs=[your_data])
```

OpenAI-Compatible Endpoint

Use existing OpenAI code with Clarifai models:

```
from openai import OpenAI

client = OpenAI(
    base_url="https://api.clarifai.com/v2/ext/openai/v1",
    api_key="your_clarifai_pat"
)

response = client.chat.completions.create(
    model="clarifai/model-url",
    messages=[{"role": "user", "content": "Your prompt"}]
)
```

Available SDKs

SDK	Installation	Best For
Python	`pip install clarifai`	Backend services, data pipelines
Node.js	`npm install clarifai`	Web applications, serverless
OpenAI-compatible	Base URL: `https://api.clarifai.com/v2/ext/openai/v1`	Easy migration
LiteLLM	See docs	Multi-provider abstraction

SDK	Installation	Best For
Vercel AI SDK	See docs	Next.js applications
REST API	Direct HTTP calls	Any language/ platform

Authentication

1. Create a PAT (Personal Access Token) at clarifai.com
2. Set environment variable: `export CLARIFAI_PAT=your_token`
3. Start making API calls

No custom services or infrastructure required.

Open-Source Model Recommendations

Recommended Primary Models (Based on Your Use Case)

Model	Parameters	License	GPU Requirement	Link
mm-poly-8b *(Clarifai)*	8B	clarifai	Standard (24GB)	mm-poly-8b ⭐
llava-1.5-7b-hf	7B	llama2	Standard (24GB)	llava-1.5-7b-hf
Llama-3.2-11B-Vision-Instruct	11B	llama3.2	Standard (24GB) or Performance (48GB)	Llama-3.2-11B-Vision-Instruct

⭐ *= Recommended Clarifai model, ready to use. Check all models at clarifai.com/explore*

Embedding Models (For RAG/Search)

Model	Parameters	License	GPU Requirement	Link
all-MiniLM-L6-v2	22M	apache-2.0	Entry (16GB) or CPU	all-MiniLM-L6-v2
bge-large-en-v1.5	335M	mit	Entry (16GB)	bge-large-en-v1.5

⭐ *= Recommended Clarifai model, ready to use. Check all models at clarifai.com/explore*

Alternative Model Options

Model	Parameters	License	GPU Requirement	Link
llava-v1.6-mistral-7b-hf	7B	apache-2.0	Standard (24GB)	llava-v1.6-mistral-7b-hf
Qwen2-VL-7B-Instruct	7B	apache-2.0	Standard (24GB)	Qwen2-VL-7B-Instruct
Phi-3-vision-128k-instruct	4.2B	mit	Entry (16GB)	Phi-3-vision-128k-instruct

⭐ *= Recommended Clarifai model, ready to use. Check all models at clarifai.com/explore*

GPU Requirements Summary

Based on the recommended models:

- **Standard (24GB)**: mm-poly-8b, llava-1.5-7b-hf, llava-v1.6-mistral-7b-hf, Qwen2-VL-7B-Instruct

- **Standard (24GB) or Performance (48GB)**: Llama-3.2-11B-Vision-Instruct
- **Entry (16GB) or CPU**: all-MiniLM-L6-v2
- **Entry (16GB)**: bge-large-en-v1.5, Phi-3-vision-128k-instruct

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Next Steps

1. Model Evaluation (Week 1-2)

- Test VLMs (starting with mm-poly-8b) against your sample data
- Benchmark accuracy, latency, and cost
- Select best model(s) for your use cases

2. Integration Development (Week 2-4)

- Integrate Clarifai SDK into your application (docs)
- Implement API calls for each use case
- Set up error handling and monitoring
- Consider Workflows for multi-model pipelines
- Use Vector Search for RAG applications

3. GPU Reservation

- Based on evaluation, determine GPU requirements
- Reserve appropriate capacity through Clarifai
- Leverage Compute Orchestration benefits

4. Production Deployment

- Deploy integrated application
- Monitor performance and adjust as needed
- Scale GPU reservation as usage grows

Resources

Resource	Link
API Documentation	https://docs.clarifai.com/
Model Library	https://clarifai.com/explore
Inference Docs	https://docs.clarifai.com/compute/inference/
Workflows	https://docs.clarifai.com/create/workflows/
AI Agents	https://docs.clarifai.com/compute/agents/
Vector Search	https://docs.clarifai.com/create/search/
Compute Orchestration	https://docs.clarifai.com/compute/overview
Python SDK	`pip install clarifai`

Clarifai manages all compute infrastructure. You focus on building your application.